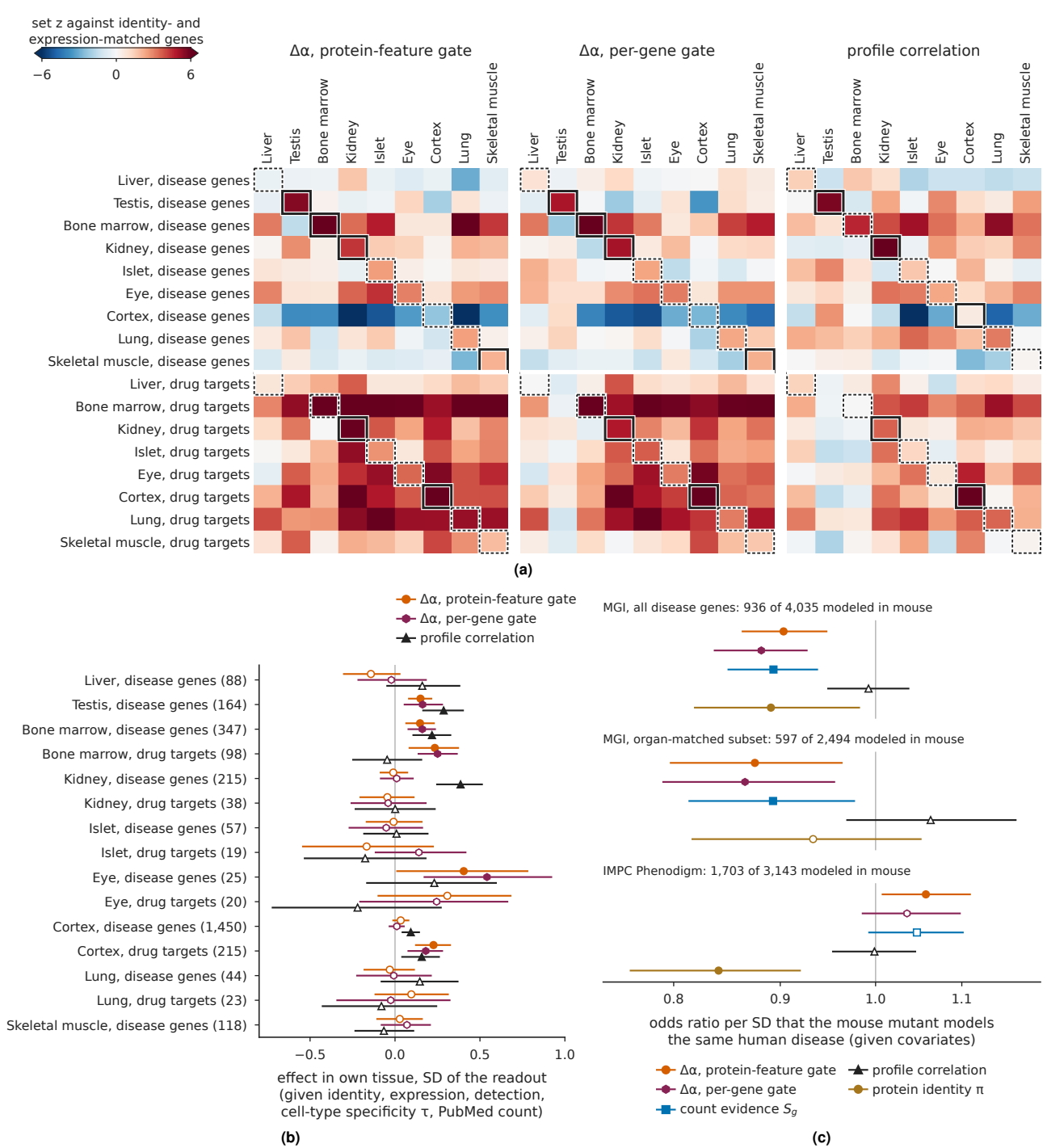

set z against identity- and expression-matched genes
−6
0
6
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Skeletal muscle
Liver, disease genes
Testis, disease genes
Bone marrow, disease genes
Kidney, disease genes
Islet, disease genes
Eye, disease genes
Cortex, disease genes
Lung, disease genes
Skeletal muscle, disease genes
Liver, drug targets
Bone marrow, drug targets
Kidney, drug targets
Islet, drug targets
Eye, drug targets
Cortex, drug targets
Lung, drug targets
Skeletal muscle, drug targets
(a)
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
Liver, disease genes (88)
Testis, disease genes (164)
Bone marrow, disease genes (347)
Bone marrow, drug targets (98)
Kidney, disease genes (215)
Kidney, drug targets (38)
Islet, disease genes (57)
Islet, drug targets (19)
Eye, disease genes (25)
Eye, drug targets (20)
Cortex, disease genes (1,450)
Cortex, drug targets (215)
Lung, disease genes (44)
Lung, drug targets (23)
Skeletal muscle, disease genes (118)
−0.5
0.0
0.5
1.0
effect in own tissue, SD of the readout
(given identity, expression, detection,
cell-type specificity τ, PubMed count)
(b)
MGI, all disease genes: 936 of 4,035 modeled in mouse
MGI, organ-matched subset: 597 of 2,494 modeled in mouse
IMPC Phenodigm: 1,703 of 3,143 modeled in mouse
0.8
0.9
1.0
1.1
odds ratio per SD that the mouse mutant models
the same human disease (given covariates)
Δα, protein-feature gate
Δα, per-gene gate
count evidence S_g
profile correlation
protein identity π
(c)